10BONDS

Livio Rating for sovereign credit

Complete calculation and research manual · Version 1.0 · 12 October 2026 · General government model · Annual horizon 2027 to 2047

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Overview

Purpose and result

This manual defines how to calculate a comparable internal sovereign credit score from 0 to 100, where higher means safer. It combines fiscal strength, refinancing protection, usable natural resources, accessible official financial support and the sovereign's postwar default record. It specifies the inputs, linear scoring endpoints, weights, evidence requirements, annual projections and publication checks.

The result is an analytical screening score, not a probability of default, an agency rating or a recommendation to buy a bond. All LR weights, endpoints, stress sizes and default-history penalties below are internal design choices. They are fully specified for reproducibility but have not been statistically calibrated or validated against subsequent sovereign defaults.

The complete formula

```
LRcore = (14S1 + 10S2 + 7S3 + 9S4 + 6S5 + 5S6 + 5S7 + 8S8 + 6S9 + 5S10 + 5S11 + 5S12 +
4S13 + 4S14 + 4S15 + 3S16) / 100


LRfinal = max(0, LRcore - H)
```

Each S is a score between 0 and 100. H is the postwar default-history deduction, between 0 and 10 points, defined on page 7. Retaining the original 16 weights preserves the 70% solvency and 30% repayment split. History is a separately disclosed overlay rather than a hidden change to those weights. Publish both LRcore and LRfinal; use LRfinal when a single LR is required.

How to apply this manual

Collect official evidence, reconcile the debt schedule, calculate each input, apply the continuous score formulas, calculate the weighted core, then deduct the history adjustment. Run the annual base and stress paths separately. If an essential input is missing, publish a bounded result or N/A, not a fabricated point score. The method is complete even when the available country evidence is not.

Reading guide

Page 2 gives the scoring table. Pages 3 to 5 define the inputs. Page 6 specifies the repayment schedule. Page 7 covers defaults since 1946. Pages 8 and 9 cover projections and stress. Page 10 provides a numerical example. Pages 11 and 12 specify sources, evidence and checks. Page 13 adapts useful corporate credit ideas without duplicating the core weights.

1. Linear scoring and weights

Use percentage-point inputs: enter 60 for debt of 60% of GDP, not 0.60. Ratios such as reserve coverage use multiples: enter 1.5 for 1.5 times. All operations retain full precision; round published scores to one decimal only at the end.

```
clip(z) = min(1, max(0, z))

L(x; a,b) = 100 × clip((b - x) / (b - a))

U(x; a,b) = 100 × clip((x - a) / (b - a))
```

L is used when lower is safer: a earns 100 and b earns 0. U is used when higher is safer: a earns 0 and b earns 100. Both are linear between their endpoints and flat outside them. Composite factors use the same linear components. Discrete counts and formula caps do not create rating buckets.

NO	PARAMETER	WEIGHT	SCORE RULE
1	Gross debt / GDP	14%	L(x; 30, 130)
2	Three year change in debt / GDP	10%	L(x; −10, 20)
3	Primary fiscal balance / GDP	7%	U(x; −5, 5)
4	Interest / government revenue	9%	L(x; 5, 30)
5	Next three years real growth	6%	U(x; −2, 6)
6	Government revenue / GDP	5%	U(x; 10, 45)
7	Net international investment position / GDP	5%	U(x; −100, 50)
8	Institutions and political stability	8%	0.4GE + 0.4RL + 0.2PS
9	Diversification and usable resources	6%	0.7D + 0.3R
10	Weighted average debt maturity	5%	U(years; 2, 12)
11	Debt due within next three years	5%	L(share %; 10, 60)
12	Next year gross financing needs / GDP	5%	L(x; 5, 30)
13	Foreign currency debt share	4%	L(share %; 0, 80)
14	Liquidity protection	4%	Regime rule on page 5
15	Market access and funding flexibility	4%	0.5A + 0.5M
16	Accessible official financial support	3%	U(coverage ratio; 0, 1)

Example: debt/GDP of 30 gives S1 = 100; 31 gives 99; 60 gives 70; 130 or more gives 0. A one-point change in S1 changes LRcore by 0.14, not by one point. The raw overall fiscal balance is still collected, but factor 2 uses debt trajectory to reduce overlap with primary balance and interest burden.

2. Fiscal and external strength

For a score dated at the end of year t, debt stocks and annual fiscal flows refer to t. Forward factors use the periods explicitly stated below. For a current snapshot, select the latest completed actual fiscal year as t; forecasts must remain visibly marked as forecasts, not relabelled actuals.

Factors 1 and 2 Debt stock and direction

S1 uses consolidated general government gross debt at nominal value divided by nominal GDP. Do not net sovereign wealth funds or private assets against gross debt. Record public guarantees, public corporations and central bank exposures separately as contingent liabilities. If only central government debt is available, label the scope limitation; do not combine its repayment schedule with general government debt and imply full coverage.

S2 uses debt/GDP in t+3 minus debt/GDP in t, measured in percentage points, not percentage growth. For example, a move from 60% to 66% gives x = 6, S2 = 46.67. Use a consistent official projection; where it ends, use the explicitly labelled scenario engine on page 8. Preserve the official series and any scenario extension separately.

Factors 3 and 4 Fiscal balance and interest

Primary balance is revenue minus non-interest expenditure, positive for surplus and negative for deficit. Use the same government perimeter, accounting basis and year for primary balance, overall balance and revenue. Interest burden is 100 times gross interest expenditure divided by total government revenue, excluding borrowing.

```
Interest burden proxy = 100 × (primary balance - overall balance) / revenue
```

The proxy may use all three inputs expressed as percentages of GDP. It is permitted only if the accounting definitions reconcile. It can represent net rather than gross interest: explicitly label a net-interest proxy and do not silently mix it with gross-interest observations. Prefer reported gross interest; if the definitions cannot be reconciled, factor 4 is N/A. Revenue of zero or less makes the ratio invalid.

Factors 5 and 6 Growth and fiscal capacity

Growth is the arithmetic average of the real GDP growth forecasts for t+1, t+2 and t+3 from one IMF WEO vintage, or an explicitly labelled scenario extension. Do not replace three forecasts with the last three actual observations. Revenue is total general government revenue including grants divided by GDP. Record grants separately because they may expire. The revenue score is a capacity proxy, not a claim that higher taxation is always better.

Factor 7 External balance sheet

NIIP equals the economy’s external financial assets minus external financial liabilities, divided by GDP, using the IMF IIP or national central bank series. It covers the whole economy, not just the government; it is not cash available to repay sovereign bonds. Never count a private foreign asset as a treasury buffer. If stock dates and GDP periods differ, disclose the mismatch and apply the data-age rules on page 11.

Primary sources: IMF WEO and Fiscal Monitor, Eurostat for consistent European government accounts, national finance ministries and central banks, and IMF IIP. The source register on page 12 provides entry links. Every country observation must additionally point to its exact series, table or report page.

3. Institutions resources and diversification

Factor 8 Institutional quality

```
S8 = 0.40 × GE + 0.40 × RL + 0.20 × PS
```

GE is Government Effectiveness, RL is Rule of Law and PS is Political Stability and Absence of Violence/Terrorism. Use the World Bank WGI absolute 0–100 scores from a single release and observation year. Do not mix absolute scores, percentile ranks and standardized estimates. This replaces the earlier political-stability-only percentile specification; the 40/40/20 blend is an internal choice.

WGI measures perceptions and has uncertainty. The World Bank warns against treating it as a definitive credit-rating or investment-risk criterion. Use this modest-weight component as contextual evidence and retain its uncertainty and vintage, not as independent proof of sovereign safety. Missing any component makes S8 N/A; no weight redistribution is allowed.

Factor 9 Economic diversity and usable resources

```
S9 = 0.70 × D + 0.30 × R

D = L(c; 0.10, 0.60)

R = U(r; 0, 10)
```

c is the UNCTAD normalized merchandise-export product concentration index, on its 0–1 scale. Lower concentration means more diversification. Use the arithmetic mean of the latest three consecutive actual annual observations. This is explicitly an export-goods proxy: it does not capture the full service economy or all sources of resilience. Do not substitute a country-level unnormalized HHI without a documented methodology change.

r is the arithmetic mean of the latest three actual annual ratios of government extractive-resource receipts to GDP, in percent. Include oil, gas and mining taxes, royalties, production-sharing receipts and dividends paid to government from extractive activities. Avoid double-counting tax and dividend transfers already consolidated. Exclude borrowing, proceeds from asset sales, private-sector export receipts and the estimated underground value of reserves.

A country with resource receipts equal to 2% of GDP receives R = 20; one at 10% or more receives R = 100. Natural resources can add at most 1.8 points to LRcore because 6% × 30% = 1.8%. A country without extractive receipts has R = 0 only when zero is verified; missing evidence is N/A. The absent bonus is not itself evidence of distress.

Required resource disclosures

Alongside the score, record oil, gas and major mineral types; the official reserve measure and year; recent production; government share of receipts; and any documented extraction, infrastructure or legal constraints. These disclosures do not create additional points. Proven reserves must not be converted into spendable fiscal wealth by multiplying them by today’s commodity price.

Resources already influence revenue, growth and debt. The small resource component intentionally adds structural context but is not independent of those factors. Commodity concentration reduces D, partially offsetting resource strength. Run the commodity stress on page 9 and disclose the overlap; do not describe the score as statistically independent or free of double-counting.

4. Repayment liquidity and support

Factors 10 to 13 Debt structure

S10 uses principal-weighted remaining years to each contractual principal payment, not the original tenor and not bond duration. Amortizing debt is split into each instalment. S11 is 100 times principal contractually due in the next 36 months divided by total outstanding principal at the scoring date. Do not include future replacement debt in the starting-date denominator. S12 is next calendar year's gross financing needs divided by that year's nominal GDP. S13 is debt denominated in currencies other than the sovereign's domestic or shared monetary-union currency, as a percentage of total debt. EUR is domestic currency for euro-area members, but foreign currency for non-euro countries.

Factor 14 Currency regime specific liquidity

For a peg or managed exchange-rate regime, or whenever foreign-currency debt exceeds 5% of total, use Q = usable official FX reserves / next 12 months economy-wide external financing needs; S14 = U(Q; 0.5, 2). The denominator is external debt principal due on remaining maturity plus the positive current-account deficit over the same period. It includes private external debt. Exclude encumbered reserves, gold and illiquid assets; do not assume all reported reserves are usable. Reserves and needs must use the same currency and date.

For a floating domestic currency or monetary-union member with foreign-currency debt at or below 5%, use Q = unrestricted central-government liquid cash and deposits / next 12 months GFN; S14 = U(Q; 0, 1). Do not treat ECB assets as a national treasury's cash. Exclude assets committed to other uses and amounts already allocated to reduce the funding gap. Publish which branch is used; these are different liquidity proxies and cross-regime comparisons require caution. If the applicable numerator is not evidenced, S14 is N/A. A verified zero denominator with no scheduled need gives S14 = 100; an unknown denominator does not.

Factor 15 Market access

```
S15 = 0.50 × A + 0.50 × M
```

A = U(q; 0.5, 1), where q is voluntary gross market debt issuance actually settled during the latest completed calendar year divided by the official issuance plan published at the start of that year. Include domestic and international bills, bonds and market loans; exclude forced placements, central bank monetary financing and official rescue loans. Do not revise the original plan retrospectively. A cancelled need or materially changed plan requires a note, not an invented ratio. If both plan and issuance are zero, A is N/A.

M = U(m; 0, 1), where m is voluntary market debt funding divided by all positive gross debt financing raised over that same year. Official loans and direct central-bank financing enter the denominator, not the numerator. If the denominator is zero, M is N/A. This funding-access proxy is deliberately not based on yields or spreads, which would make portfolio selection circular. It does not measure investor concentration; disclose that concentration separately.

Factor 16 Accessible official support

S16 = U(F / GFN; 0, 1). F is approved, undrawn official financing legally available for sovereign general financing within the next 12 months, with required conditions satisfied at the observation date. Include evidenced IMF, EU, ESM and bilateral facilities. Exclude mere membership, hoped-for rescues, already-disbursed loans, unavailable tranches, central-bank purchase programmes and ringfenced project grants. No facility means F = 0 only after verification. If GFN = 0, score 100 when F > 0 and otherwise 0. Conditions or amounts that cannot be verified mean N/A, not automatic support.

5. The debt repayment schedule

An average maturity cannot reveal a refinancing wall. Build a principal and interest ledger covering every outstanding security and loan, including short bills, domestic debt and official loans. An international-bond list alone is insufficient for a sovereign repayment score.

FIELD	REQUIRED DEFINITION
Identity and perimeter	Instrument or loan ID, issuer, currency, guarantee status and government subsector.
Principal	Outstanding amount, each amortization date and amount, final maturity and any indexation.
Interest	Coupon or contractual loan rate, reset formula, payment dates and reference-rate assumptions.
Optionality	Borrower and creditor options, contractual first put date, and treatment of calls.
Evidence	Official source URL, table/page, publication date, retrieval date and reported coverage.

Use contractual maturities unless the holder can require earlier repayment, in which case use the earliest exercisable put date. Do not assume a borrower will call its own debt. For floating rates use the official forecast if available; otherwise hold the latest observed contractual reference rate constant and label the assumption. For indexed principal apply the specified official inflation path. A perpetual debt stock with no finite repayment date prevents a finite whole-stock average maturity: publish the finite-maturity subportfolio statistic and mark S10 N/A unless the official whole-stock convention is explicitly available.

```
Average maturity = sum(principal payment × years to payment) / outstanding principal

Three year share = 100 × principal due within 36 months / outstanding principal

Net funding need = principal due + cash interest - cash primary balance + K

GFN = max(0, net funding need)
```

K is net other cash requirements: financial-asset purchases, recognized guarantee payments and other cash outlays minus non-debt financing receipts. Keep debt issuance and cash-buffer withdrawals outside K. A positive primary balance reduces funding need. If using accrual fiscal balances, include a sourced accrual-to-cash bridge; without it, label GFN an approximation, not an exact cash forecast.

Reconciliation and no double counting

Principal in the ledger must reconcile to the matching reported debt stock. Report the uncovered residual explicitly. Require at least 95% value coverage and an official maturity allocation for the remaining amount before publishing a point score for S10, S11 or schedule-derived S12. Never spread unknown redemptions evenly to make the schedule balance. If only official aggregate maturity and GFN series are available, score those as reported but label detailed wall analysis incomplete.

Calculate GFN before cash withdrawals, market borrowing or official loan disbursements. Calculate the residual funding gap after allocating each source exactly once. An official loan disbursed today becomes debt and possibly treasury cash; it is no longer undrawn support. Do not count the same amount as cash and undrawn support. Publish annual maturities through 2050, all later maturities in their actual years, and monthly cash needs for the next 24 months where official data permit. Annual adequacy alone cannot prove that an intrayear payment can be met.

6. Sovereign defaults since the Second World War

Record the number of distinct default episodes beginning from 1 January 1946 through the information cutoff. Also report episodes already ongoing at that date separately. A default record is evidence about repayment behaviour, but the number alone is not a calibrated future default probability.

What constitutes one episode

Include failure to pay sovereign debt principal or interest after any contractual grace period; repudiation; or a distressed restructuring that imposes creditor loss or less favourable terms to avoid nonpayment. Include domestic and foreign-currency debt and official-creditor debt, with separate labels. Exclude ordinary voluntary market exchanges and general supplier arrears from the scored count; disclose the latter separately. A debt-service suspension agreed without distress is not automatically a default.

Count an episode once from its initial qualifying event until obligations in that episode have been cured or restructured and performance has resumed. Multiple creditor groups, Paris Club rounds and exchanges resolving the same unresolved crisis are not additional country episodes. Concurrent qualifying events form one episode. Count a new event after cure as another episode. If exact dates are unavailable, a run of positive annual default observations separated by at least one full zero-default year can be used only as a labelled annual-data proxy; do not present it as an exact event count.

Required country history record

Publish total post-1945 episode count, start and cure dates, domestic/external and private/official creditor classification, affected debt, reason for classification, and direct official evidence. Report the last cure date, years since cure, current unresolved default status and historical coverage start. A state formed after 1945 starts at sovereignty; identify inherited liabilities and predecessor defaults separately, adding an episode only where the new state legally assumed the defaulted obligation. Do not assign all predecessor defaults automatically.

History deduction with linear ageing

```
E = sum over episodes of max(0.25, 1 - a / 40)

H = min(10, 2 × E)
```

a is years since the documented cure date, using days / 365.25 and floored at zero. An ongoing episode has a = 0. A cured episode retains weight 1 at cure, 0.75 after 10 years, 0.50 after 20 years and 0.25 from 30 years onward. Thus one episode deducts 2, 1.5, 1 or 0.5 LR points respectively. Repetition accumulates up to a maximum deduction of 10 points. Include a pre-1946 episode still ongoing in 1946 once in E and disclose it separately from the count of new postwar defaults.

These penalty sizes are internal, uncalibrated choices. They preserve a lasting but modest historical penalty and a larger recent penalty. If history coverage is incomplete, report a verified minimum count and a range for LRfinal; do not turn “not found” into zero defaults. Unknown cure dates imply an episode weight range of 0.25 to 1, not an invented date.

Sources and current distress

Use the Bank of Canada–Bank of England database for coverage from 1960, cross-checked with IMF reports, national debt-office records and official creditor notices. IMF restructuring research helps for the 1950s but counts restructurings, not necessarily distinct defaults. Investigate 1946–1959 separately; neither dataset alone establishes a complete postwar count. Any unresolved default, debt-service moratorium or material debt arrears must carry a prominent CURRENT DEFAULT flag. A high numerical LR must not override that status or qualify the sovereign as performing debt.

7. Annual LR from 2027 to 2047

Produce a separate end-year score for every year 2027–2047, using only information available at the stated research cutoff. A score for 2047 needs macro inputs through 2050 because debt trajectory and growth look three years ahead. These are conditional scenario scores, not verified future ratings.

Official forecasts first and explicit extensions second

Use the latest released IMF WEO vintage and the latest official debt-management strategy available by the cutoff. Respect each source's actual, estimate and forecast labels. Where the official forecast ends, hold its final real growth rate, GDP-deflator inflation, revenue/GDP and primary-balance/GDP ratios constant. Use GDP-deflator inflation, not CPI, for nominal GDP. Hold NIIP/GDP, institutional scores, export concentration and resource receipts/GDP constant after their last available values. Label all such carried values as scenario assumptions; never attribute the 2047 extension to the IMF.

```
GDP(t+1) = GDP(t) × (1 + real growth / 100) × (1 + deflator inflation / 100)


Debt close = debt open + new borrowing - principal repaid + valuation and other
adjustments
```

Take existing-debt payments from the ledger. New borrowing fills positive net funding need after approved official disbursements and planned cash withdrawals. Default base cash policy: no discretionary asset sales or buffer withdrawals; apply a negative funding need to cash accumulation. Recognize every approved official disbursement as a loan with its own interest and repayment schedule. No approved drawdown schedule means no assumed future disbursement. Support availability ends at contractual expiry, without assumed renewal.

New borrowing rules

Use the official financing strategy for currency, instrument mix and tenor. After its forecast ends, freeze the latest official planned mix; if there is no plan, use the latest completed year's actual issuance mix and contractual terms. Use the latest actual weighted issuance yield per currency and tenor as the base refinancing rate unless an official projected funding-cost series exists. Do not use an agency rating to invent a rate. If neither issuance records nor a usable official plan exists, future debt-structure scores are N/A.

Annual simplification: new borrowing settles on 31 December, with its first coupon in the following year and principal repaid at its specified tenor; a one-year bill repays the following December. This removes circular within-year interest calculations but understates intrayear funding complexity. Where a dated official issuance plan exists, use it instead and calculate actual within-year interest. Retain all newly issued debt and its eventual maturities beyond 2050 when computing remaining maturity. Do not delete long debt at the modelling boundary.

After the official projection ends, hold base exchange rates constant. Preserve source-backed stock-flow adjustments within official forecasts; set later unannounced adjustments to zero and mark this assumption. If the ledger projection differs from official total debt, disclose a reconciliation residual instead of forcing unknown issuance or maturities. Carry S15 forward at its

last measured value in the base path, explicitly as an access assumption, not a forecast. Age history penalties only conditional on no further defaults; never infer zero future defaults as an empirical prediction.

For the FX branch of factor 14, extend the last official reserve/GDP and external-financing-needs/GDP ratios unchanged after their forecast horizon, clearly labelling this as a liquidity-ratio scenario rather than an external-sector model. In stresses hold their base foreign-currency amounts, revaluing them at the stressed exchange rate; this omits current-account feedback. For the treasury-cash branch, update cash with modelled surpluses, documented withdrawals and disbursements. Keep unresolved defaults unresolved until an evidenced cure; do not invent a future cure date.

8. Stress testing and score interpretation

Calculate a base path and the following separate stresses, all starting in the first model year after the latest actual year. Recompute fiscal flows, debt, refinancing needs and score components consistently. Do not assign probabilities to these scenarios.

STRESS	DETERMINISTIC ASSUMPTION
Growth	Real growth 2 percentage points below base for two years; then base rates resume from the lower GDP level.
Refinancing cost	New borrowing rates and floating-rate resets 3 percentage points above base from the stress start onward; fixed existing coupons unchanged.
Currency	One-time 20% rise in the local-currency price of every foreign currency, then hold the new level. Revalue foreign debt, service and usable FX assets consistently.
Resources	Government extractive receipts 30% below base for two years. Deduct lost receipts from revenue and primary balance once; update the rolling resource ratio.
Contingent liabilities	One-time cash payment of 5% of starting-year GDP, funded as a new requirement; do not also add the same amount directly to debt.
Combined	Apply all five together without adding the same fiscal loss twice. No exchange-rate shock to debt denominated in the domestic or shared currency.

For the growth stress, maintain base revenue/GDP but keep nominal primary expenditure at its base amount; recalculate the primary balance. The resource stress is additional and subtracts only the specified lost receipts. Apply currency conversion to original-currency cash flows before aggregation. Hold domestic inflation at base in these mechanical stresses; this is a limitation, not a macroeconomic forecast. Keep NIIP/GDP and governance at base unless a separately evidenced alternative is specified.

For base and stress paths, calculate financing need before financing. Show how much funding is assumed to be available and the residual gap. A debt path that mechanically assumes all borrowing succeeds is a financing requirement, not proof of market access. Do not change S15 purely because issuance was assumed; disclose that its carried value is conditional. In a separate access-loss test, set new voluntary market funding to zero for one year and show unmet payments, without treating the resulting borrowing shortfall as debt reduction or an improvement in solvency.

How to interpret the output

Publish LRcore, history deduction H, LRfinal, solvency subscore and repayment subscore, plus all raw inputs and scenario labels. Solvency subscore equals the weighted sum of factors 1–9 divided by 0.70; repayment subscore equals the weighted sum of factors 10–16 divided by 0.30. Each block therefore remains on a 0–100 scale.

Do not translate LR 75 into a specific agency rating or probability of default. That requires an independent historical calibration with observation dates, out-of-sample tests and a defined default horizon. Do not use 100 minus LR as default probability. Similarly, higher LR is not a guarantee of repayment, lower volatility or a higher expected investment return.

For a EUR bond, show currency, legal issuer, guarantee, seniority, contractual maturity and current agency rating separately. A country LR does not automatically rate a state-owned company or

every security issued by the country. Portfolio loss probabilities require issuer-level default probabilities, dependence and recovery assumptions; this manual supplies none of those probabilities.

9. Worked example and arithmetic tests

This is an illustrative country, not an estimate for any real sovereign. All raw inputs below are hypothetical. It uses the FX-reserve branch for factor 14 and complete historical coverage.

NO	RAW INPUT	SCORE	LR CONTRIBUTION
1	60% debt/GDP	70.0000	9.8000
2	0 pp three year debt change	66.6667	6.6667
3	1% primary surplus	60.0000	4.2000
4	10% interest/revenue	80.0000	7.2000
5	3% growth	62.5000	3.7500
6	30% revenue/GDP	57.1429	2.8571
7	−25% NIIP/GDP	50.0000	2.5000
8	GE 80 RL 75 PS 60	74.0000	5.9200
9	c 0.30 and r 2%	48.0000	2.8800
10	8 years remaining maturity	60.0000	3.0000
11	25% due within three years	70.0000	3.5000
12	12% next year GFN/GDP	72.0000	3.6000
13	20% foreign currency share	75.0000	3.0000
14	Reserves cover needs 1.5 times	66.6667	2.6667
15	Issuance plan 90% met and market share 80%	80.0000	3.2000
16	Official undrawn support covers 25% GFN	25.0000	0.7500

Using unrounded values, LRcore = 65.490476, displayed as 65.5. Factor 2: 100 × (20 − 0) / 30 = 66.6667. Factor 9: D = 60, R = 20, so S9 = 48.

Assume two cured default episodes, one 10 years ago and one 40 years ago. E = 0.75 + 0.25 = 1; H = 2. LRfinal = 63.490476, displayed as 63.5. The postwar count is 2 and the last cure was 10 years ago. There is no current-default flag.

Minimum automated checks

Weights must sum to 100%; block weights must sum to 70% and 30%. Every score must lie in [0,100]. Test both endpoints, their midpoint and out-of-range values. Confirm debt 31% scores 99, a 0 pp debt change scores 66.6667, and a 20-year-old episode deducts one LR point. Recalculate totals from unrounded values, and verify a worsening raw input cannot improve its own score with all other inputs held fixed.

10. Evidence missing data and output records

Required evidence for every input

Store country ISO code; parameter number; raw value and unit; observation date or fiscal year; actual/estimate/official forecast/scenario status; government perimeter; source institution; document title and publication vintage; exact table, page or series identifier; direct URL; access date; and all transformations. Save the original download and its SHA-256 hash. For derived figures, preserve numerator and denominator with their separate evidence.

Use the latest actual year available by the cutoff for each observed non-forward input. If fiscal inputs refer to different years or perimeters, do not combine them in a ratio. Financial stocks and schedules older than 18 months, or annual governance, NIIP, export and resource data older than 36 months, are stale for a current snapshot: publish their age and withhold an unqualified point LR. In a forward scenario, carrying a value is permitted only with an explicit assumption label; it does not make the data current.

Missing information is not zero

```
Core lower bound = sum of known weight × score

Core upper bound = core lower bound + 100 × sum of missing weights
```

Weights in these equations are fractions of one. Do not renormalize the known weights. A composite with missing components is missing in full. Publish data coverage as the percentage of total weight with valid inputs. Only publish a fully supported point LR when all 16 components and the history adjustment are supported, and the debt-schedule coverage condition is met where schedule-derived measures are used.

For uncertain history use Hmin from documented episodes and their minimum supported penalties; Hmax is 10 if additional episodes cannot be ruled out. Then LRfinal lies between max(0, core lower − Hmax) and max(0, core upper − Hmin). Clearly label bounds as missing-information limits, not statistical confidence intervals. With complete core but incomplete history, core can be published independently.

Required annual output table

One record per country, year and scenario: LRcore; H; LRfinal or bounds; solvency and repayment subscores; all 16 raw inputs, scores and weighted contributions; postwar episode count or verified minimum; last cure date; current-default flag; debt due by year; GFN; planned financing; funding gap; currency-regime branch; data coverage; and actual/forecast/scenario labels. Retain every year from 2027 to 2047, without implying precision beyond the assumptions.

Operational workflow

1. Freeze cutoff and source vintages. 2. Collect and reconcile debt, macro, institutions, resources, facilities and history. 3. Calculate the current snapshot. 4. Build the debt and macro path through at least 2050. 5. Calculate every annual component and history adjustment. 6. Run the stresses. 7.

Apply all validation tests. 8. Export the evidence table, debt ledger, history ledger, assumption register and score table alongside the rating summary.

Update after a new major macro release, debt-management plan, official programme decision, default event or material revision. Never replace old results silently: preserve version, date and a bridge explaining each change. A reproducible implementation must use these formulas directly; Monte Carlo is neither required nor appropriate for calculating this score.

11. Official source register and application limits

The links below identify official source families and methodology. They are starting points, not substitutes for the exact country observation link. Archive the table actually used. Source institutions do not endorse the LR weights, thresholds or deductions.

IMF WEO and Fiscal Monitor. Factors 1–6 and macro forecasts. Use the full database and vintage appendix; verify the actual-versus-estimate boundary. https://data.imf.org/Datasets/WEO

Eurostat government finance statistics. European debt, balances, revenue and consistent government perimeter. https://ec.europa.eu/eurostat/web/government-finance-statistics/database

IMF IIP. Factor 7 external assets and liabilities; cross-check the national central bank. https://data.imf.org/Datasets/IIP

World Bank WGI. Factor 8 absolute scores and usage advisory; the LR blend is internal. https://www.worldbank.org/en/publication/worldwide-governance-indicators

UNCTAD Data Centre. Factor 9 merchandise export concentration; finance ministries supply extractive receipts. https://unctadstat.unctad.org/datacentre/reportInfo/US.ConcentStructIndices

IMF sovereign risk framework. Debt sustainability, financing needs and contingent-liability context; not LR thresholds. https://www.imf.org/en/publications/dsa/sovereign-risk-and-debt-sustainability-analysis-for-market-access-countries

World Bank debt statistics. QPSD government debt structure and QEDS external debt remaining-maturity data. https://www.worldbank.org/en/programs/debt-statistics

National official institutions. Factors 10–16: DMO redemption schedules and issuance plans; treasury cash; central-bank reserves; signed programme and facility decisions. Use each country's official site and exact document, not an aggregator. Country-specific official document URL required in every data record.

Bank of Canada and Bank of England. Default database and definitions from 1960; use the latest release available at cutoff. Reconcile creditor observations into country episodes. https://www.bankofcanada.ca/2023/07/technical-report-124/

IMF historical restructuring research. Evidence for 1950–2010; restructurings are not automatically distinct defaults. Verify 1946–1949 and all uncovered periods from national archives and official creditor records. https://www.imf.org/en/publications/wp/issues/2016/12/31/sovereign-debt-restructurings-1950-2010-literature-survey-data-and-stylized-facts-26190

Final publication gate

Release a score only with its information cutoff, model version, source coverage and status flags. A default history is complete only when the full applicable postwar period has been checked. Do not claim zero defaults from a database that starts in 1960. Do not claim a 2047 forecast is official when it is a constant-assumption extension. Country ratings, probabilities of default and investment decisions remain separate outputs requiring their own evidence and validation.

12. Ideas adapted from corporate credit analysis

Corporate credit ratios offer useful questions about cash, repayment and financial discipline. Governments have no directly comparable EBITDA, and taxing powers, monetary arrangements and official financing change the interpretation. The adaptations below are mandatory supporting diagnostics, not additional weighted factors in version 1.0. Publish them beside LR; do not add their scores to LRcore or deduct them again from LRfinal.

Twenty four month liquidity coverage

```
Q24 = (C + F24) / (GFN year 1 + GFN year 2)

Liquidity diagnostic = U(Q24; 0, 2)
```

C is unrestricted treasury cash available at the scoring date. F24 is approved undrawn financing usable for general funding within 24 months, meeting the same eligibility rules as factor 16. Include each amount once, only when available before the payment it is intended to cover. Exclude assumed future market issuance, central-bank reserves not available to the treasury, and loan proceeds already in C. The denominator uses GFN before these funding sources. Coverage of 0, 0.5, 1, 1.5 and 2 times scores 0, 25, 50, 75 and 100. A verified zero funding need scores 100. Missing either yearly GFN means N/A. The cumulative 24-month ratio must be accompanied by the earlier monthly-gap check: later cash cannot repay an earlier bill.

Five year fiscal cash generation record

```
Consistency diagnostic = 100 × positive primary balance years / 5
```

Use five consecutive completed actual years of primary balance on one government perimeter. A strictly positive balance counts as one; zero or a deficit counts as zero. Scores are 0, 20, 40, 60, 80 or 100 because the count is discrete, not because thresholds are imposed. Missing any year makes the diagnostic N/A. Show the five balances and their mean beside the score so a tiny surplus does not conceal weak fiscal capacity. This imports the cash-generation consistency idea without calling public investment an avoidable corporate capex item.

Fiscal policy credibility

```
e = mean absolute error of five original overall balance forecasts

Credibility diagnostic = L(e; 0, 5)
```

For each of the same five completed years, compare the original approved government budget’s overall balance/GDP forecast with the latest official actual outcome, in percentage points. Take each absolute difference, then their arithmetic mean. Error of 0, 1, 2.5 and 5 points scores 100, 80, 50 and 0. Use the original budget, not a revised in-year forecast. Missing an original forecast or actual means N/A. Disclose perimeter changes and exceptional shocks; the mechanical measure assesses predictability, not intent or blame.

Ideas already covered and ideas not transferred

Corporate leverage maps broadly to debt/GDP and its trajectory; interest coverage to interest/revenue; earnings trend to growth and primary balance; business and industry quality to governance, export concentration and resource exposure. These are conceptual parallels, not numerical equivalences. Importing their corporate weights would duplicate existing factors. Do not derive sovereign business quality from an agency rating and then claim LR is an independent rating.

The corporate macro block is better used as named scenarios than six unsupported judgements. Growth, energy, interest-rate and currency shocks are specified on page 9. Document country-specific war, trade, China-competition and pandemic exposures qualitatively; add a numerical scenario only with explicit transmission assumptions. Do not award arbitrary 0–100 scores to those narratives. The threshold ladders in the corporate table are not imported: all continuous LR inputs retain the linear rules in this manual.